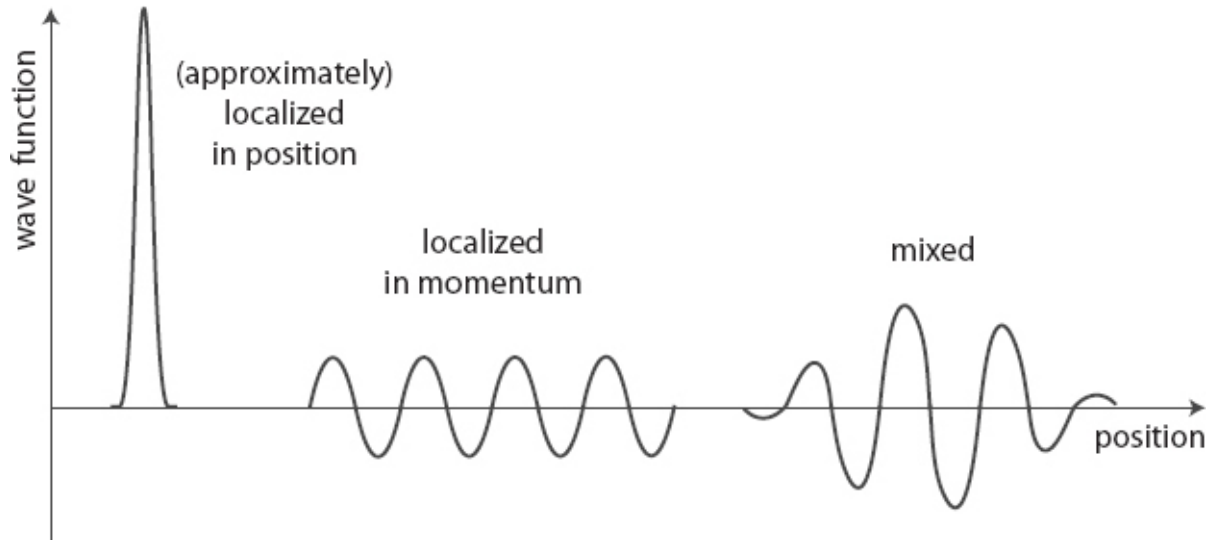


We see the basic dilemma. If we try to localize a wave function in space, its momentum becomes more and more spread out, and if we try to limit it to one fixed wavelength (and therefore momentum) it becomes more spread out in position. That's the uncertainty principle. It's not that we can't *know* both quantities at the same time; it's just a fact about how wave functions work that if position is concentrated near some location, momentum is completely undetermined, and vice versa. The old-fashioned classical properties called *position* and *momentum* aren't quantities with actual values, they're possible measurement outcomes.

People sometimes refer to the uncertainty principle in everyday contexts, outside of the equation-filled language of physics texts. So it's important to emphasize what the principle does *not* say. It's not an assertion that "everything is uncertain." Either position or momentum could be certain in an appropriate quantum state; they just can't be certain at the same time.

And the uncertainty principle doesn't say we necessarily disturb a system when we measure it. If a particle has a definite momentum, we can go ahead and measure that without changing it at all. The point is that there are no states for which both position and momentum are simultaneously definite. The uncertainty principle is a statement about the nature of quantum states and their relationship to observable quantities, not a statement about the physical act of measurement.

Finally, the principle is not a statement about limitations on our knowledge of the system. We can know the quantum state exactly, and that's all there is to know about it; we still can't predict the results of all